PORTFOLIO MANAGEMENT PROCESS

Quantitative portfolio managers pride themselves on making rational investment decisions, constructing efficient investment portfolios, and maximizing investor utility. In fact, this is the central theme of modern portfolio theory (MPT). Portfolio managers following this course of action will seek to maximize return for a specified quantity of risk (variance).

This optimization is formulated as follows:

$$\begin{aligned} &Max \quad w'r \\ &s.t. \quad\; w'Cw \le \sigma_p^{2*} \\ &\qquad\quad \textstyle\sum w = 1 \end{aligned} \tag{10.1}$$

where w is the vector of weights, r is the vector of expected returns, C is the covariance matrix, and σ_p^{2*} is the targeted or maximum level of risk.

This optimization can also be formulated as the dual of Equation 10.1 where the goal is to minimize risk for a targeted return r^*. This optimization is formulated as:

$$\begin{aligned} &Min \quad w'Cw \\ &s.t. \quad\; w'r \le r^* \\ &\qquad\quad \textstyle\sum w = 1 \end{aligned} \tag{10.2}$$

The set of all solutions to the portfolio optimization problem above results in the set of all efficient portfolios and comprises the efficient frontier. This is the set of all portfolios with highest return for a given level of risk or the lowest risk for a specified return. Proceeding, to avoid confusion, we use the term efficient investment frontier (EIF) to denote the set of optimal investment portfolios (Markowitz and Sharpe) and the efficient trading frontier (ETF) to denote the set of optimal trading strategies (Almgren and Chriss).

The set of all optimal portfolios can also be found using Lagrange multipliers as follows:

$$\begin{aligned} &Max \quad w'r - \lambda \cdot w'Cw \\ &s.t. \quad\; \textstyle\sum w = 1 \end{aligned} \tag{10.3}$$

where λ denotes the investor's risk appetite (e.g., level of risk aversion). Solving for all values of $\lambda \ge 0$ provides us with the set of all efficient portfolios.